7 A 2.0 kg block slides down a plane that is inclined at 25° with the horizontal, at a uniform speed of 2.0 m s^{-1}. What is the rate of frictional force doing work on the block?

A 17 W **B** 22 W **C** 36 W **D** 46 W

A

$$F_{friction} = (2)(9.81)(\sin 25°) = 8.292\,\text{N (negative)}$$

$$\text{Power} = \frac{\text{WD}}{\text{t}} = F\frac{d}{t} = (F_{friction})(\text{Velocity}) = 16.6\ \text{W (negative)}$$

8 A pendulum bob of mass 1.27 kg is supported by a string so that the radius of its